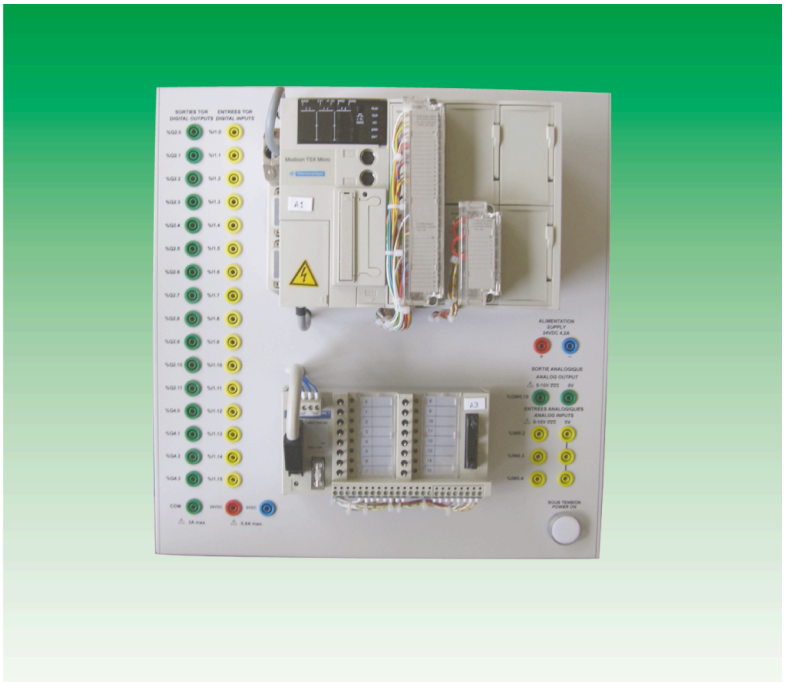
TSX Micro on support with sockets

Study and programming in PLC language



Schneider Electric

1. Presentation of the equipment

■ The "TSX Micro on support with sockets" consists in a PLC TSX Micro installed on a support including sockets for various connections. It is designed for people who want to learn programming a PLC in language "List / Ladder, Litteral and Grafcet".

1.1 Composition

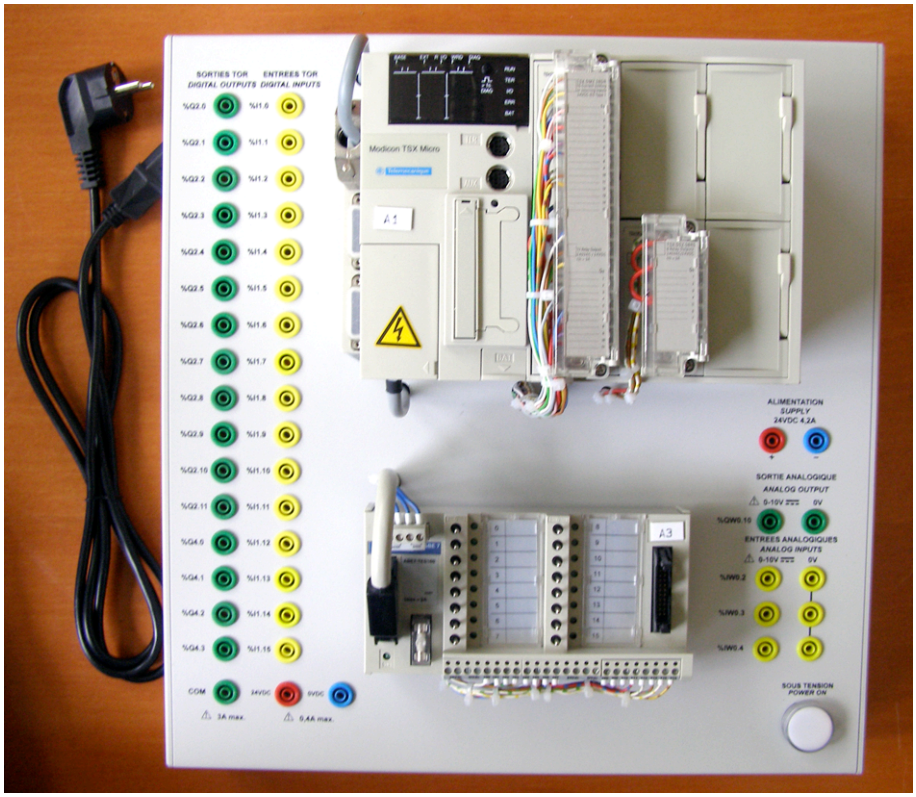
■ This assembly is designed for easy and safe connection to operative parts or equipments.

■ A printing next to each socket represents the inputs/outputs.

■ Various LED give information on the operation or the status of the PLC inputs and outputs.

■ A socket for simulation plugged in parallel to the inputs allows testing of a program or an application without operative part.

■ A built-in power supply "Phaseo" 24 Vdc 4,2 A can supply the actuators following their characteristics.



1.2 Data

■ Mechanical data

☐ Enclosure : height: 250 mm, width: 400 mm, depth: 350 mm

☐ Weight: 5 kg approximately

■ Electrical data

☐ Voltage: 230V single-phase ± 10 %

☐ Power absorbed: 30 VA

☐ Frequency: 50 Hz ± 5 %

☐ Current: 16 A, connection by plug 2Ph + Earth

☐ Conventional short-circuit current: 3 kA.

☐ Rated surge withstand voltage: 2.5 kV

☐ Class of protection to electric shocks: II

☐ The electrical network must be fitted upstream of the equipment with an RCD (Residual Current Device) with a sensitivity of ≤ 30 mA class AC (not supplied).

■ Technical data of the main components:

☐ PLC

It is a TSX3721001 with power supply 100 - 240 Vac including :

– Internal RAM memory of 20Kwords
– a slot for a memory extension card
– a slot for a communication card
– a terminal socket (TER)
– a socket for dialogue with the operator (AUX)
– a back-up battery TSXPLP01.
– a module 16 inputs 24Vdc / 12 relay outputs
– a module 8 relay outputs

☐ A power supply "Phaseo" 24 V 4,2 A maximum, reference ABL7RE2403.

☐ A simulation socket, reference ABE7TS160.

☐ 16 yellow safety sockets with the printing: %I1.0 to %I1.15 for connection of inputs.

☐ 16 vertes safety sockets with the printing: %Q2.0 to %Q2.11 and %Q4.0 to %Q4.3 for connection of outputs.

☐ 3 safety sockets under the inputs/outputs sockets, with the following printings:

– COM to connect the common points of the outputs in green.
– 24Vdc 0.4 A maxi: 24 Vdc supplied by the power supply of the TSX Micro in red
– 0 Vdc: 0 V supplied by the power supply of the TSX Micro in blue.

☐ 2 red and blue safety sockets under the "phaseo supply", with the printing: power supply 24 Vdc 3A.

☐ A white LED indicator "SOUS TENSION" ("UNDER VOLTAGE").

1.3 Main functions

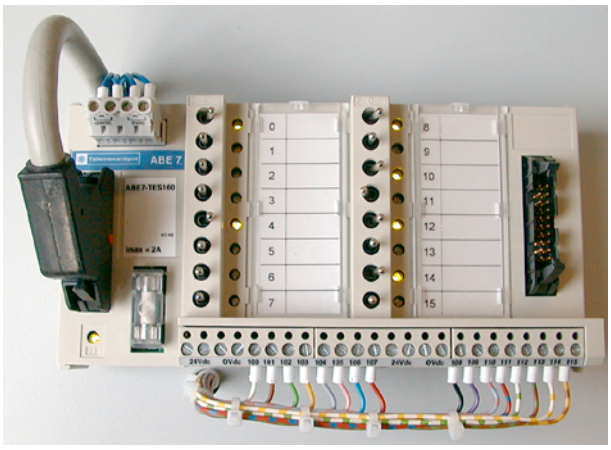
The "TSX micro on support with sockets" can be used with or without operative part:

■ Without operative part, simulating a program:

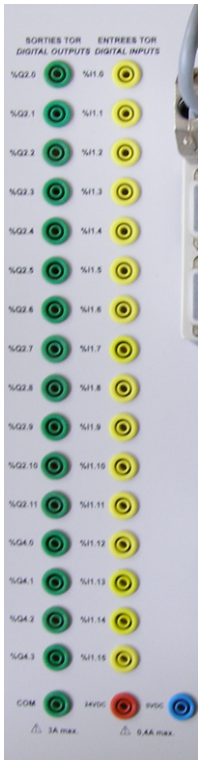
☐ Manipulations can be carried out on the PLC: creation or loading of programs, simulation with the switches of the test module and the indicators of the display unit located on the front panel of the TSX Micro.

■ With operative part:

☐ Exercises can be carried out on the PLC: creation or loading of existing programs, implementation of the operative part, simulation of a program operation with the switches and the indicators located on the front panel of the TSX Micro.







2. Training course

2.1 Presentation of the training course

■ The "TSX Micro on support with sockets" is used to learn:

☐ The various languages, Contacts or Ladder, Grafcet, Litteral,

☐ Programming in combinatorial or in sequential,

☐ The various function blocks (delay devices, comparators, operations, registers, etc.),

☐ Work on words, arithmetic operations,

☐ The event tasks,

☐ communication through the socket "terminal auxiliary" with various extension units (for instance: dialogue terminal).

■ It is intended for the following educational training cycles:

	Professional	High School	University Engineering school
Automated systems – industrial computing	❑	❑	❑
Electrical engineering		❑	

☐ The hands-on exercises that can be carried out using the "TSX Micro on support with sockets" are different depending on the operative parts connected to it.

☐ An example of "traffic management" that can be performed without operative part is given in the following pages.

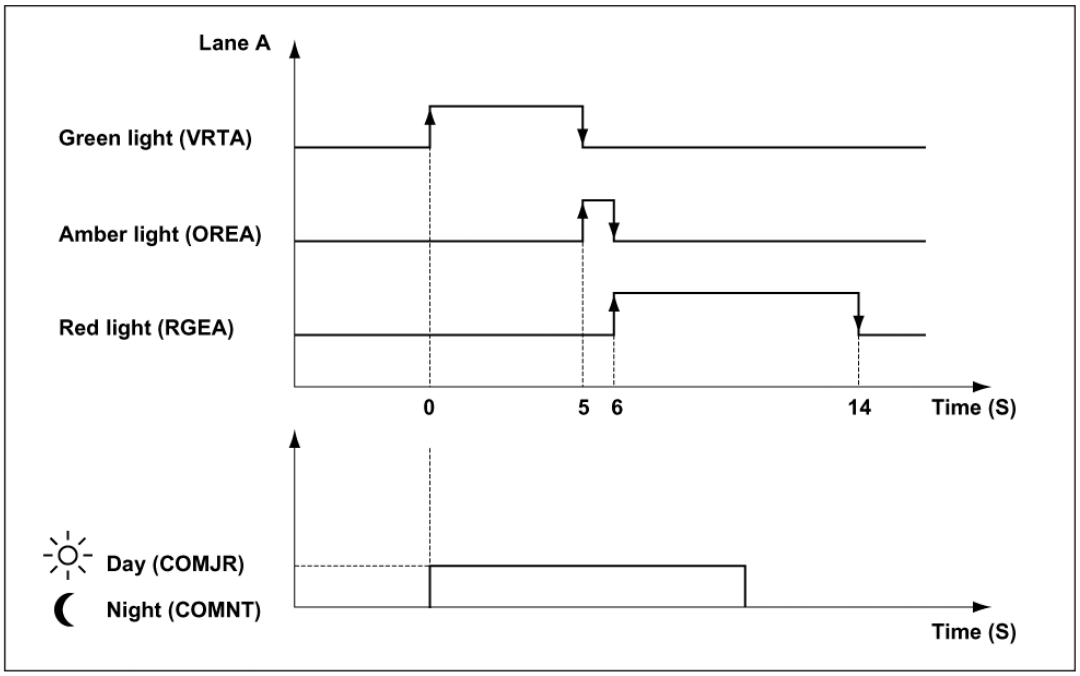
2.2 Training example

Hands-on exercises - Study of lane A in "day-time" operation

Variables used

Description	Sensors / Actuators	PLC addresses	Ti	TB	PRESET
"Day" position selector	COMJR	% I 1.7	T0	1s	5
Red light lane A	RGEA	% Q 2.0	T1	100ms	10
Amber light lane A	OREA	% Q 2.1	T2	1s	8
Green light lane A	VRTA	% Q 2.2			

Chronogram



Exercise 3 - Study of lane A in "day-time" operation

(The time delay)

Introduction

■ **Exercise aim**

❏ Study the TIME DELAY using the GRAFCET.

■ **Knowledge required**

❏ Elementary functions of the GRAFCET

See Reference Manual 1, Description of the PL7 Software.

Exercise

■ **Situation**

❏ The module is stopped.

❏ When the selector (COMJR) is placed in the DAY position :

- the green light in lane A comes on,
- 5 s. later: the green light goes out, then the amber light comes on,
- 1 s. later : the amber light goes out, then the red light comes on,
- 8 s. later: the red light goes out.

❏ If (COMJR) is still in the DAY position, the cycle is resumed.

❏ If (COMJR) is not in the DAY position, i.e. in the NIGHT position, the module moves to STOP.

❏ The aim is to create the installation GRAFCET and programming of this GRAFCET and the TIME DELAYS.

■ **Work required**

1 - Draw the functional GRAFCET of the installation.

2 - Determine from it the operational GRAFCET.

3 - Write this GRAFCET in PL7 MICRO language.

4 - Use the time delay relays (% TMi).

5 - Configure the time delay relays before commissioning.

6 - CONFIGURATION mode: Time delay no. / Time Base (TB) / Preselection / Setting option.

7 - Program and test operation.

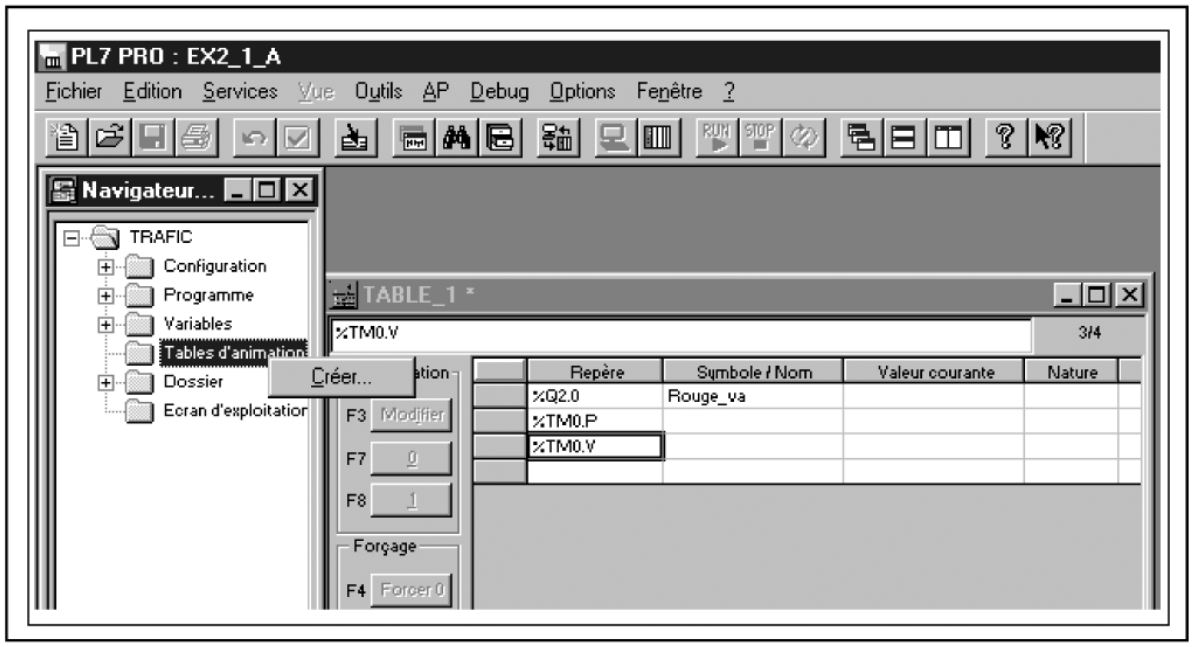
8 - Using the dynamic view mode, check evolution of current value (%Ti.V) of the time delay relays: %T0.V - %T1.V - %T2.V.

(see creation of an animation table to view a PLC table).

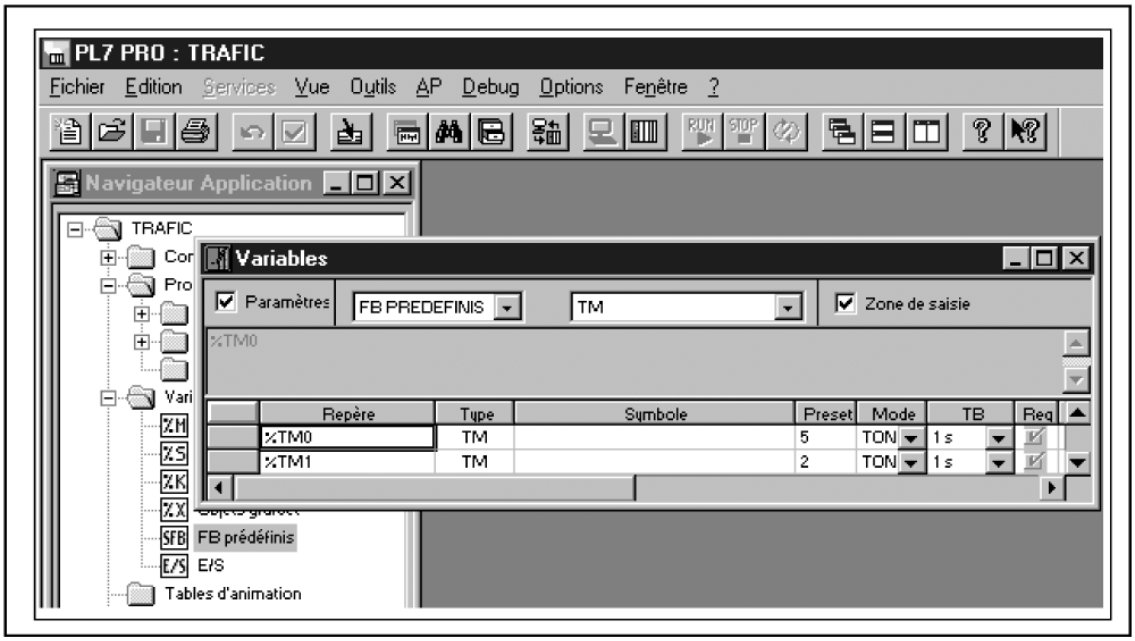
9 - Try to modify in this mode the preselection value of %T0 and its time base (TB): what happens ?

(see «Reference Manual 2, Predefined fonction blocks»)

■ Creation of an "animation table" to view dynamic object values:



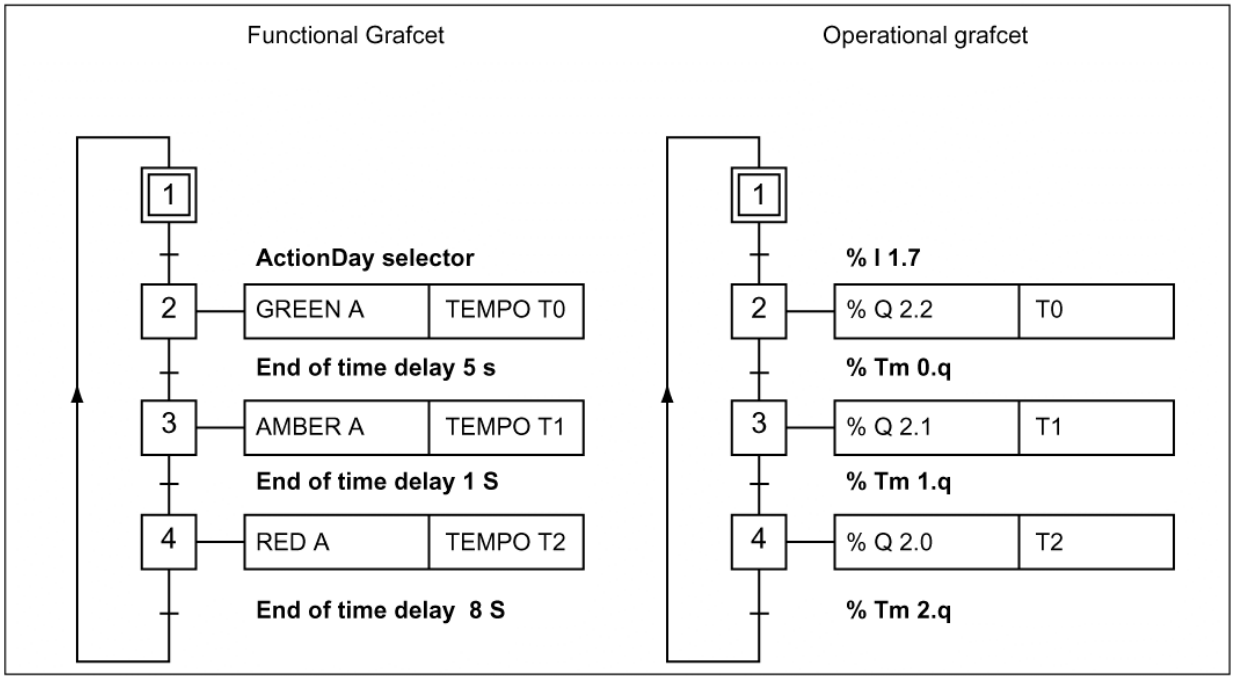
■ Configuration example of predefined function blocks



Corrections for exercise 3

■ Programming

The functional and operational grafcets are given below:



❏ See the print-out of programs below:

■ Comments

❏ The current value of a time delay relay changes by one unit every time base.

❏ It is possible to modify the time base (TB) if the PLC is on STOP. However, the time delay preselection value can be modified in RUN mode via an animation table. The time delay relay setting mode is left at "Yes".

Corrections for exercise 3 (continued)

■ Exercise 3 - program print-out

CHART - PAGE 0



CHART - PAGE0 %X(1)->%X(2)

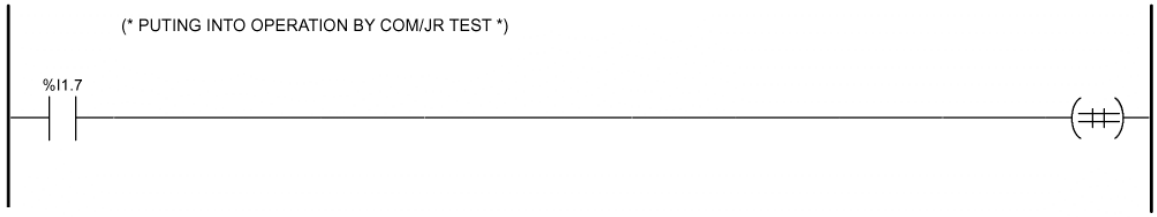


CHART - PAGE0 %X(2)->%X(3)



CHART - PAGE0 %X(3)->%X(4)

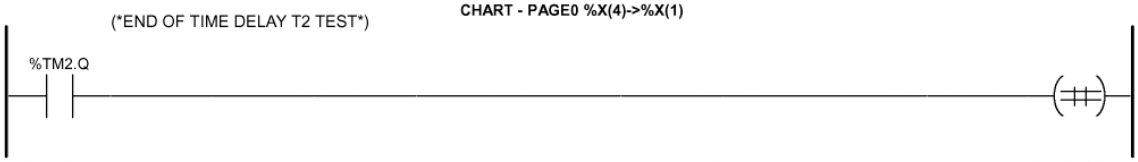


Corrections for exercise 3 (continued)

CHART - PAGE0

CHART - PAGE0 %X(4)->%X(1)

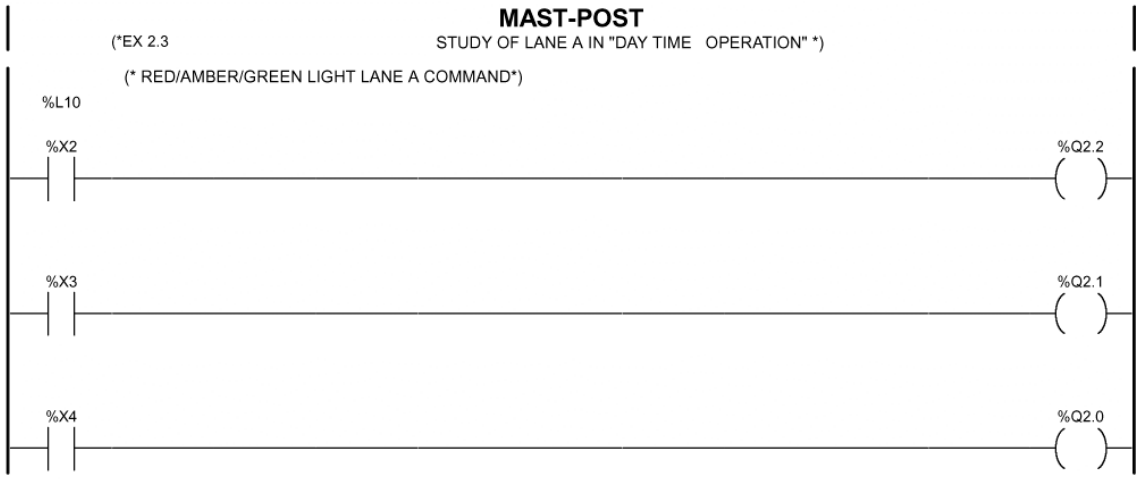
(*END OF TIME DELAY T2 TEST*)

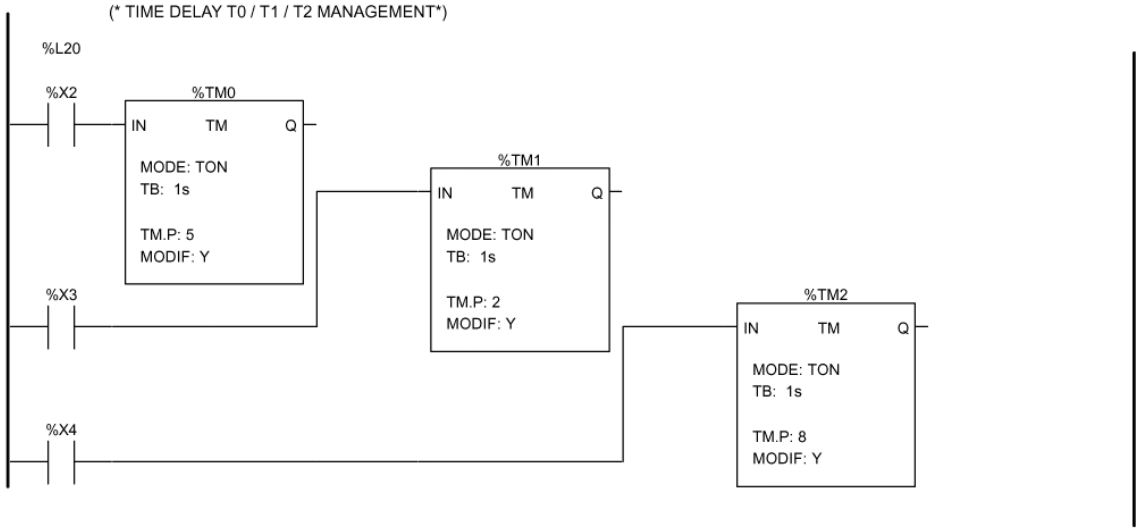


Liste de Variables utilisées dans le rung :

REPERE	SYMBOLE	COMMENTAIRE
%TM2.Q		

Corrections for exercise 3 (continued)





3. Items supplied with the equipment

3.1 Devices supplied (Reference MD1AD110)

■ A TSX3721 equipped with:

☐ a power supply 24Vdc 3A for inputs and actuators,

☐ a socket for simulation, reference ABE7TS160,

☐ a cord for mains with a plug 2Ph + Earth 10/16 A for connection to the network,

☐ a white indicator.

3.2 Devices not supplied

■ Programming software for TSX Micro in language "Ladder, Litteral and Grafcet" with its cable:

– PL7 Micro, PL7 Junior or PL7 Pro

■ the micro computer,

■ the extensions, their connecting cable and their software,

■ the cords for safety sockets.

3.3 Documentation

■ A technical manual reference MD1AD111EN

4. Contact details

Schneider Electric France
Activité Didactique 3F
35 rue Joseph Monier
CS 30323 92506 Rueil Malmaison - France

http://www.schneider-electric.com/educationalequipments
e-mail: FR-didactic@fr.schneider-electric.com